

# Lecture 2. Deep Learning for Continuous Time Models and Structural Estimation

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# Background: deep learning for continuous-time models

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- Lecture 1: discrete-time HA models with aggregate shocks; the distribution as a state (“master equation”) solved with deep learning (DeepHAM)
- Many dynamic models are set in continuous time: HJB + KFE for HA models (Achdou et al '22), macro-finance (Brunnermeier-Sannikov), search and matching
- With heterogeneity + aggregate shocks, equilibrium = high-dimensional PDE
- Deep learning solves such PDEs globally: neural nets as function approximators, PDE residuals as loss, economics-guided sampling and training
- Furthermore, solve models with structural parameters as pseudo state variables  $\Rightarrow$  deep learning based structural estimation
- This lecture is based on Payne, Rebei & Yang (2026), “Deep Learning for Search and Matching Models” (DeepSAM)

# Introduction

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- **Heterogeneity** and **aggregate shocks** are important in markets with **search frictions**. (e.g. labor and financial markets).
- Most search and matching (SAM) models with heterogeneous agents study:
  1. Deterministic steady state (e.g. Shimer-Smith '00),
  2. Aggregate fluctuations, but impose restrictions to eliminate distribution from state space (e.g. “block recursivity” in Menzio-Shi '11, Lise-Robin '17; Lagos-Rocheteau '09).
- We present SAM models as **high-dim. PDEs** with **distribution** & **agg. shocks** as states ...and develop a new deep learning method, DeepSAM, to **solve** them globally.
- We extend DeepSAM for **internal calibration with SMM** in efficient computation time.

# DeepSAM: Payne, Rebei & Yang (2026)

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- Develop DeepSAM and apply to “canonical” search models with aggregate shocks:
  1. Shimer-Smith/Mortensen-Pissarides model with two-sided heterogeneity.
  2. Lise-Robin on-the-job search (OJS) model with worker bargaining power.
  3. Duffie-Garleanu-Pederson OTC model with asset and investor heterogeneity.
- High accuracy + efficient compute time for both solution and estimation.
- This talk: study unemployment and wage dynamics during business cycles and crises:
  1. Distribution feedback most important when aggregate shocks are asymmetric.
  2. Low-type worker wages are more procyclical.
  3. Low-type workers benefit more from longer expansions (“Okun’s hypothesis”).
  4. Block recursive model assumptions amplify IRFs.

# Literature

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- Deep learning for incomplete-market heterogeneous agent models (HAM): [Lecture 1](#)
  - [DeepSAM: search and matching \(SAM\) models.](#)

|     | Distribution                                              | Distribution impact on decisions         |
|-----|-----------------------------------------------------------|------------------------------------------|
| HAM | Asset wealth and income                                   | Via aggregate prices                     |
| SAM | Type (productivity) of agents<br>in two sides of matching | Via matching process<br>with other types |

- Search model with business cycle (e.g. Shimer '05, Menzio-Shi '11, Lise-Robin '17.)
  - [DeepSAM: keep distribution in the state vector.](#)
- Integrate deep learning based solution methods with calibration and estimation (e.g. Chen et al '26, Kase et al '26, Friedl et al '23)
  - [DeepSAM: standard internal calibration practice for quantitative macro.](#)

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# Shimer-Smith Model with Aggregate Shocks

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- Continuous time, infinite horizon environment.
- **Workers**  $x \in [0, 1]$  with exog density  $g_t^w(x)$ ; **Firms**  $y \in [0, 1]$  with  $g_t^f(y)$  by free entry:
  - Unmatched: unemployed workers get benefit  $b$ ; vacant firms pay vacancy cost  $c$ .
  - Matched: type  $x$  worker and type  $y$  firm produce output  $z_t f(x, y)$ .
  - $z_t$ : follows continuous time Markov Chain (can be generalized).
  - Firms make entry decision and then draw a type  $y$  from uniform distribution  $[0, 1]$ . More
  - Firm exit: exogenous exit at rate  $\varsigma(z)$ ; fraction  $\sigma(z, \tilde{z})$  of firms exit when  $z$  jumps to  $\tilde{z}$ .  
(Exiting firms/workers compensated; calibrated s.t. **no endogenous exit**.) General case
- **Meet randomly** at rate  $m(\mathcal{U}_t, \mathcal{V}_t)$ ,  $\mathcal{U}_t$  is total unemployment,  $\mathcal{V}_t$  is total vacancies.
- Upon meeting, agents choose whether to accept the match:
  - Match surplus  $S_t(x, y)$  divided by **Nash bargaining**: worker gets fraction  $\beta$ .
  - Acceptance decision  $\alpha_t(x, y) = \mathbb{1}\{S_t(x, y) \geq 0\}$ . Exogenous dissolve rate  $\delta(x, y, z)$ .
- Equilibrium object:  $g_t(x, y)$  “density” of matches  $\Rightarrow$  unemployed  $g_t^u(x)$ , vacant  $g_t^v(y)$ .

# Recursive Equilibrium: Agent Problems

- E.g. worker idiosyncratic state =  $x$ , Aggregate states =  $(z, g(x, y))$ .  
Let  $dg/dt = \check{\mu}^g(x, y, z, g)$  denote agents' belief about the evolution of  $g$ .
- Hamilton-Jacobi-Bellman equation (HJBE) for unemployed worker's value  $V^u(x, z, g)$ :

$$\begin{aligned} \rho V^u(x, z, g) = & b + \frac{m(z, g)}{U(z, g)} \int \underbrace{\overbrace{\alpha(x, \tilde{y}, z, g)}^{\text{acceptance decision}} \overbrace{(V^e(x, \tilde{y}, z, g) - V^u(x, z, g))}^{\text{employed value}}}_{\text{change of value conditional on match}} \frac{g^v(\tilde{y})}{V(z, g)} d\tilde{y} \\ & + \sum_{\tilde{z}} \lambda_{z\tilde{z}} (V^u(x, \tilde{z}, g^+) - V^u(x, z, g)) + \underbrace{D_g V^u(x, z, g)}_{\text{Frechet derivative: how change of } g \text{ affects } V} \cdot \underbrace{\check{\mu}^g}_{\text{Belief about } g \text{ evolution}} \end{aligned}$$

where  $g^+ := (1 - \sigma(z, \tilde{z}))g$ : post-jump distribution (firm exit at  $z$  jumps).

# Recursive Equilibrium: Agent Problems

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+ HJBEs for employed workers  $V^e$ , vacant firms  $V^v$ , and producing firms  $V^p$  More.

# Recursive Equilibrium: Distribution Evolution and Definition

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Given  $\alpha$  choices, **evolution of  $g_t(x, y)$**  is given by Kolmogorov forward equation (KFE):

$$dg_t(x, y) = \mu^g(x, y, z_t, g_t) dt - \underbrace{\sigma(z_t, \tilde{z}) g_t(x, y) dN_t(z_t; \tilde{z})}_{\text{Jump: exogenous firm exit when } z_t \text{ jumps to } \tilde{z}}$$

$$\mu^g(x, y, z, g) = \underbrace{-(\delta(x, y, z) + s(z)) g_t(x, y)}_{\text{Outflow: breakup of matches \& firm exit}} + \underbrace{\frac{m(z, g)}{\mathcal{U}(z, g)\mathcal{V}(z, g)} \alpha(x, y, z, g) g_t^u(x) g_t^v(y)}_{\text{Inflow: creation of new matches}}$$

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A (recursive) equilibrium is functions  $\{V^u, V^e, V^v, V^p, w, \alpha, g^f\}$  of state variables  $(z, g)$  such that:

1. Given beliefs about the evolution of  $g_t$ ,  $(V^u, V^e, V^v, V^p, \alpha)$  solve the HJB equations,
2. The division of surplus satisfies Nash bargaining,
3.  $g^f$  satisfies the free entry condition details, and
4. Agent beliefs about the evolution of  $g_t$  are consistent:  $\check{\mu}^g = \mu^g$ .

# Recursive Characterization For Equilibrium Surplus

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- Surplus of match  $S(x, y, z, g) \equiv V^p(x, y, z, g) - V^v(y, z, g) + V^e(x, y, z, g) - V^u(x, z, g)$
- We characterize equilibrium with master equation for surplus: C.f. block recursive

$$\begin{aligned} \rho S(x, y, z, g) = & z f(x, y) - \delta(x, y, z) S(x, y, z, g) \\ & + c - (1 - \beta) \frac{m(z, g)}{\mathcal{V}(z, g; S)} \int \alpha(\tilde{x}, y, z, g) S(\tilde{x}, y, z, g) \frac{g^u(\tilde{x})}{\mathcal{U}(z, g)} d\tilde{x} \\ & - b - \beta \frac{m(z, g)}{\mathcal{U}(z, g)} \int \alpha(x, \tilde{y}, z, g) S(x, \tilde{y}, z, g) \frac{g^v(\tilde{y})}{\mathcal{V}(z, g; S)} d\tilde{y} \\ & + \sum_{\tilde{z}} \lambda_{z\tilde{z}} (S(x, y, \tilde{z}, (1 - \sigma(z, \tilde{z}))g) - S(x, y, z, g)) + D_g S \cdot \mu^g(z, g) \end{aligned}$$

- Kolmogorov forward equation (KFE):  $dg_t(x, y) = \mu^g dt - \sigma(z_t, \tilde{z}) g_t(x, y) dN_t(z_t; \tilde{z}),$

$$\mu^g(x, y, z, g) = -(\delta(x, y, z) + \varsigma(z)) g(x, y) + \frac{m(z, g)}{\mathcal{U}(z, g) \mathcal{V}(z, g)} \alpha(x, y, z, g) g^v(y) g^u(x)$$

- High-dim PDEs with distribution in state: hard to solve with conventional methods.

# Finite Type Approximation

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- Approximate  $g(x, y)$  on finite types:  $x \in \mathcal{X} = \{x_1, \dots, x_{n_x}\}$ ,  $y \in \mathcal{Y} = \{y_1, \dots, y_{n_y}\}$ .  
(Other approaches of discretization: projection and finite population [More](#))
- Finite state approximation  $\Rightarrow$  analytical (approximate) KFE:  $g \approx \{g_{ij}\}_{i \leq n_x, j \leq n_y}$
- Approximated master equation for surplus:

$$0 = \mathcal{L}^S S(x, y, z, g) = -(\rho + \delta(x, y, z))S(x, y, z, g) + zf(x, y) + c - b$$

$$- (1 - \beta) \frac{m(z, g)}{\mathcal{V}(z, g)} \frac{1}{n_x} \sum_{i=1}^{n_x} \alpha(\tilde{x}_i, y, z, g) S(\tilde{x}_i, y, z, g) \frac{g^u(\tilde{x}_i)}{\mathcal{U}(z, g)}$$

$$- \beta \frac{m(z, g)}{\mathcal{U}(z, g)} \frac{1}{n_y} \sum_{j=1}^{n_y} \alpha(x, \tilde{y}_j, z, g) S(x, \tilde{y}_j, z, g) \frac{g^v(\tilde{y}_j)}{\mathcal{V}(z, g)}$$

$$+ \sum_{\tilde{z}} \lambda_{z\tilde{z}} (S(x, y, \tilde{z}, g^+) - S(x, y, z, g)) + \sum_{i=1}^{n_x} \sum_{j=1}^{n_y} \partial_{g_{ij}} S(x, y, z, \{g_{ij}\}_{i,j}) \mu^g(\tilde{x}_i, \tilde{y}_j, z, g)$$

where  $g_{ij}^+ = (1 - \sigma(z, \tilde{z}))g_{ij}$ ;  $\mu^g$  from discretized KFE.

# Discrete Choice Approximation

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- In the original model, acceptance is discrete:  $\alpha(x, y, z, g) = \mathbb{1}\{S(x, y, z, g) \geq 0\}$
- Discrete choice  $\alpha \Rightarrow S(x, y, z, g)$  non-smooth at some  $g$  (a viscosity solution)  
 $\Rightarrow$  sharp decision boundaries make gradient-based NN training less stable.
- Baseline: keep master equation well defined & NN training efficient  $\Rightarrow$  approximate:

$$\alpha(x, y, z, g) = \frac{1}{1 + e^{-\xi S(x, y, z, g)}}$$

Interpretation: logit choice model with utility shocks  $\sim$  extreme value distribution  
(micro-foundation as in Caliendo et al. '19;  $\xi \rightarrow \infty \Rightarrow$  discrete choice  $\alpha$ ).

- We can also solve the original discrete choice formulation, via “homotopy”:
  1. Train on a sequence of smooth problems with increasing curvature  $\xi_1 < \dots < \xi_N$ , warm-starting each stage from the previous solution;
  2. Final stage: switch to the exact discrete  $\alpha$ , initialized from the  $\xi_N$  solution.
- Delivers accurate solutions to the discrete choice problem ( $\xi = 42$  already  $\approx$  discrete),  
...but  $\sim 10\times$  compute time: smooth  $\alpha$  preferable if logit interpretation acceptable.

Results

# DeepSAM Algorithm for Solving the Master Equation

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1. Approximate surplus by neural network  $S(x, y, z, g) \approx \hat{S}(x, y, z, g; \Theta)$ . Function form
2. Start with initial parameter guess  $\Theta^0$ . At iteration  $n$  with  $\Theta^n$ :
  - 2.1 Generate  $K$  sample points,  $Q^n = \{(x_k, y_k, z_k, \{g_{ij,k}\}_{i \leq n_x, j \leq n_y})\}_{k \leq K}$ .
  - 2.2 Calculate the average mean squared error of surplus master equation on sample points:

$$L(\Theta^n, Q^n) := \frac{1}{K} \sum_{k \leq K} \left| \mathcal{L}^S \hat{S}(x_k, y_k, z_k, \{g_{ij,k}\}_{i \leq n_x, j \leq n_y}) \right|^2$$

- 2.3 Update NN parameters with stochastic gradient descent (SGD) or variants:

$$\Theta^{n+1} = \Theta^n - \zeta^n \nabla_{\Theta} L(\Theta^n, Q^n)$$

- 2.4 Repeat until  $L(\Theta^n, Q^n) \leq \epsilon$  with precision threshold  $\epsilon$ .

3. Once  $S$  is solved, we have  $\alpha$  and can solve for worker and firm value functions.

Tutorial (Colab notebooks): [Tutorials/Tutorial\\_DeepSAM in the course repository](#)

## Comment: Some Technical Details

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- Q. How do we choose where to sample? Three-phase sampling & training scheme:
  - **Phase 1**: draw distributions “between” steady states (random mixtures of steady states for different fixed  $z$ ); train until the loss is small.
  - **Phase 2**: move to simulation based (“ergodic”) sampling under the current  $\hat{S}$ , with a training/validation split, and periodic resimulation as  $\hat{S}$  improves.
  - **Phase 3**: report out-of-sample loss on a freshly simulated evaluation set. [More](#)
- Q. Why global solution? Non-linear approximation across broad state space.
- Q. Which equilibrium concept?
  - We solve for a rational expectations equilibrium  
 $\Rightarrow$  requires solving the master equation where  $\mu^g = \check{\mu}^g$  on and off equilibrium path.
  - Different to “reinforcement learning” where agents optimize in response to simulations  
 $\Rightarrow \mu^g \neq \check{\mu}^g$  does not necessarily hold, especially off equilibrium path. [More](#)

Neural networks do not train in seconds.

So how can we choose structural parameters to match data?

# DeepSAM for Internal Calibration with Simulated Method of Moments

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- So far, we have solved:

$$0 = \mathcal{L}^S S(x, y, z, \underline{g})$$

- For internal calibration, we include economic parameters  $\Psi$  as pseudo states and solve extended master equation:

$$0 = \mathcal{L}^S S(x, y, z, \underline{g}, \Psi)$$

- Simulate the model under different structural parameter vectors  $\{\Psi_I\}_I$  and ...fit a surrogate function mapping structural parameters to simulated moments.
- Use surrogate function to find the parameters that match data moments.

# Extensions

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- On-the-job search with a parameterized bargaining protocol nesting: [More](#)
  - Bertrand competition between firms (Postel-Vinay and Robin, 2002),
  - sequential auctions (Cahuc, Postel-Vinay and Robin, 2006), and
  - Nash bargaining over incremental surplus from moving (Pissarides, 1994).
- Match specific idiosyncratic productivity shocks. [More](#)
- Endogenous separation. [More](#)
- Variations on free entry and exit. [More](#)
- Agent type switching. [More](#)
- Asset trade rather than worker-firm production (e.g., OTC search markets). [More](#)
- Non-transferable utility. [More](#)

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Methodology

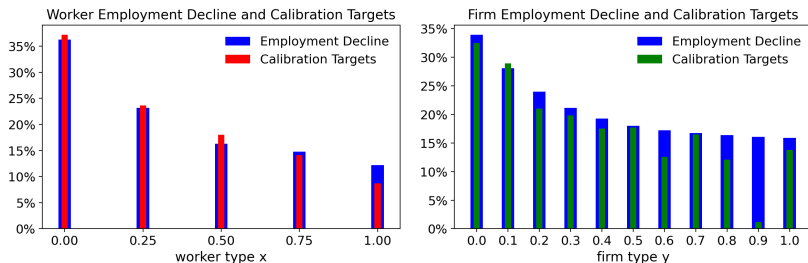
Application 1: Crisis Shock in Shimer-Smith (2000)

Application 2: Business Cycles in Lise-Robin (2017) With Worker Bargaining Power

Conclusion

# Crisis Shock in Shimer-Smith (2000)

- We test the solution method for the baseline model  
...with a recession where the job separation rate significantly increases.
- We calibrate the crisis separation rate  $\delta_D(x, y)$  to match the heterogeneous employment declines during COVID-19 for different workers/firms (Cajner et al., 2020). [More](#)



- Worker and firm types:  $(n_x, n_y) = (5, 11) \Rightarrow 58$  dimensional PDE for  $S(x, y, z, g)$

# Numerical Accuracy of Solution Technique

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Technically, our method achieves high numerical accuracy using a number of measures:

- **Small numerical error.** Use DeepSAM to solve the problem (58 dimensional PDE), and compute loss on simulated paths not used in training. Evaluation RMSE:  $1.7 \times 10^{-3}$  (training RMSE:  $9.2 \times 10^{-4}$ ). [More](#)
- **Verification on models with known solution.** Use DeepSAM to solve model without aggregate shocks (57 dimensional PDE) and obtain steady state solution. Compare to steady state solution from conventional methods. RMSE Difference:  $2.4 \times 10^{-3}$ . [More](#)

# Q1. Does Distribution Feedback Matter? Evidence From COVID-19

- Aggregate dynamics **with** and **without** distribution feedback to agent decisions:

Full dynamics:  $\mu_t^g(x, y) = -(\delta(x, y, z_t) + \varsigma(z_t))g_t(x, y) + \frac{m_t(z, g_t)}{\mathcal{U}_t \mathcal{V}_t} \alpha(x, y, z_t, g_t) g_t^u(x) g_t^v(y)$

No distribution feedback:  $\mu_t^g(x, y) = -(\delta(x, y, z_t) + \varsigma(z_t))g_t(x, y) + \frac{m_t(z, g_t)}{\mathcal{U}_t \mathcal{V}_t} \alpha(x, y, z_t, g^{\text{ergodic}}) g_t^u(x) g_t^v(y)$

(Both share the same exogenous exit jumps at  $z_t$  changes;  $g^{\text{ergodic}}$ : ergodic mean of  $g_t$ .)

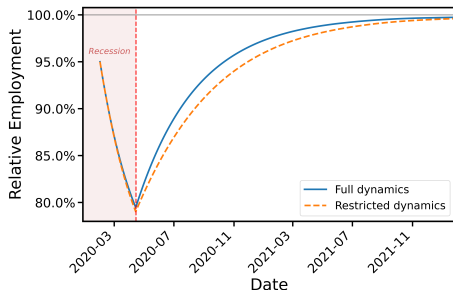
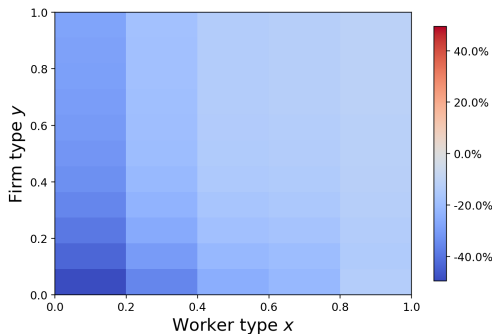


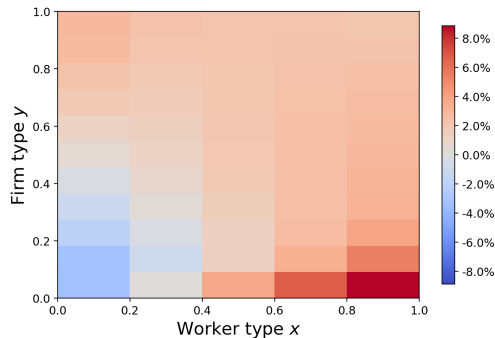
Figure: Employment change after the COVID-19 shock

# A1. Mechanism: Asymmetric Shock Makes Agents “Less Picky”

Intuition: COVID-19 leads to relatively more low-type unemployed workers and firms  
⇒ workers and firms are less willing to wait for a good match.



(a) Distribution difference: after COVID-19 shock relative to the ergodic mean.



(b) Acceptance difference: after COVID-19 shock relative to the ergodic mean.

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# Solving and Estimating OJS Model

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- Lise-Robin (2017) with worker bargaining power  $\beta > 0$  [Model details](#)
- Discretization:  $(n_x, n_y) = (7, 8) \Rightarrow S(x, y, z, g)$  is 59-dimensional.

# Solving and Estimating OJS Model

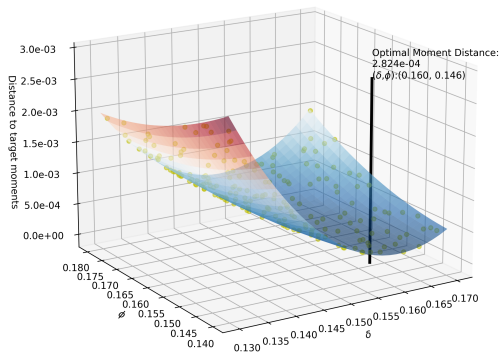
- Lise-Robin (2017) with worker bargaining power  $\beta > 0$  Model details
- Discretization:  $(n_x, n_y) = (7, 8) \Rightarrow S(x, y, z, g)$  is 59-dimensional.
- Internal calibration  $\{\beta, \kappa, c, \delta, \phi, \eta, b\}$ : solve model over economic parameter space, simulate across 10,000 parameter combinations for simulated method of moments.

|           | Solving Model For a Parameter Set | Solving Model with Structural Parameters as Pseudo-states | Training Surrogate Model | Simulated Method of Moments | Entire Estimation |
|-----------|-----------------------------------|-----------------------------------------------------------|--------------------------|-----------------------------|-------------------|
| RMSE Loss | $9.97 \times 10^{-4}$             | $2.9 \times 10^{-3}$                                      | $7.82 \times 10^{-4}$    | $1.24 \times 10^{-4}$       | -                 |
| Time      | 40min                             | 2h 47min                                                  | 23min                    | 0.4min                      | 3h 10min          |

| Moments | $\mathbb{E}[U]$ | $\mathbb{E}[V]$ | $\mathbb{E}[EE]$ | $\mathbb{E}[UE]$ | $\mathbb{E}[EU]$ |
|---------|-----------------|-----------------|------------------|------------------|------------------|
| Data    | 0.058           | 0.037           | 0.025            | 0.421            | 0.025            |
| Model   | 0.057           | 0.036           | 0.026            | 0.448            | 0.027            |

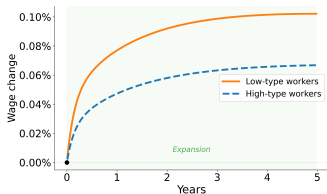
# Estimation of OJS Model: Visualization in 2D



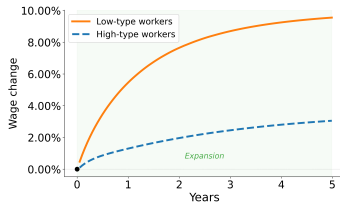
Distance to target moments over separation rate  $\delta$  and OJS intensity  $\phi$ .

## Q2. Are wage dynamics heterogeneous across distribution? A2. Yes

- In Lise-Robin: “wages cannot be solved for exactly... need to solve worker values where the distribution of workers across jobs is a state variable.”
- DeepSAM can solve wage dynamics with rich heterogeneity.
- Low-type worker wages more procyclical — under both protocols  $\chi = 0$  and  $\chi = 1$ .



(a) IRF to positive shocks ( $\chi = 0$ )



(b) IRF to positive shocks ( $\chi = 1$ )

**Figure:** Wage change after  $z$  shocks for  $\chi = 0$ ,  $\chi = 1$ . Low/middle/high: lowest, middle, highest of  $n_x = 7$  worker types.

### Q3. Who benefits more over a longer expansion? A3. Low Types

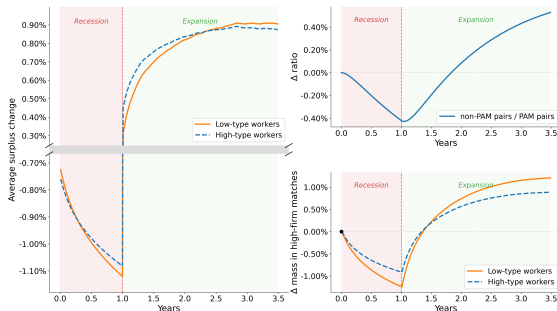


Figure: Left:  $\Delta$  avg. surplus by worker type. Top right: non-PAM/PAM mass ratio. Bottom right:  $\Delta$  employment at firms with  $y > 1/2$ .

- Low-type workers overtake high types in surplus growth  $\sim 2$  years into the expansion.
- Mechanism: sorting weakens over time in expansions — high-type workers become scarce, so high-type firms become less picky and hire more low-type workers.

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# Conclusion

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- We develop an integrated global solution and estimation method, DeepSAM, to search and matching models with heterogeneity and aggregate shocks.
- We apply DeepSAM to three general setups in labor and financial search models (without simplification assumptions). [OTC model details](#)
- The method is accurate, solves new variables (e.g. wage), and generates novel economic insights.
- A foundational tool for a large literature with more applications:
  - Richer models in labor, financial, and money search, combined with rich micro data.
  - Spatial, trade, and network models with aggregate uncertainty (similar math structure).

# The Agenda of “AI for Economics”: Two Generations of Work

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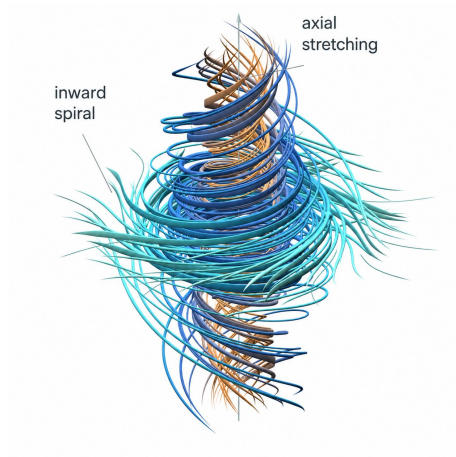
- First generation: “[proof of concept](#)” of using AI to solve economic models.
  - Duarte (2018), Azinovic, Gaegauf, Scheidegger (2019), Maliar, Maliar, Winant (2019), Fernández-Villaverde, Hurtado, Nuño (2019), Han, Yang, E (2021), Kahou, Fernández-Villaverde, Perla, Sood (2021), Valaitis and Villa (2021), Gopalakrishna (2022), Huang (2022), Gu, Lauriere, Merkel, Payne (2023), Zhong (2023), among others.
  - All papers solve similar [classical models](#) in this generation.
- Second generation: apply AI to [long-standing difficult questions in various literature](#).
  - [Climate economics](#): Folini, Friedl, Kubler, Scheidegger (2021), Friedl, Kubler, Scheidegger, Usui (2023), Barnett, Brock, Hansen, Hu, Huang (2023), Sun (2023).
  - [Labor and financial markets](#): Payne, Rebei, and Yang (2026).
  - [Asset pricing](#): Zhang (2022), Azinovic-Zemlicka (2023), Gopalakrishna, Gu, Payne (2024).
  - [HANK models](#): F-V, Marbet, Nuño, Rachedi (2022), Kase, Melosi, Rottner (2022).
  - [Network models](#): Carvalho, Covarrubias, Nuño (2025).
  - Many other fields: trade, optimal policy, information, structural estimation, etc.

# Summary

---

- Three elements of using deep learning to solve economic models:
  1. Parameterize the functions of interest as neural networks, can impose economic restrictions in parameterization
  2. Specify the objective
  3. Choose where to evaluate the objective (sampling)
- As is happening in other disciplines ([Nobel Prize Physics 2024](#), [Chemistry 2024](#), [Navier-Stokes Solution 2026](#)), AI may become the new generation of model solvers in economics.

# Thank You!



# Approximate $S$ by Neural Network (Feed Forward, Fully Connected)

---

- Let  $\omega = (x, y, z, g)$ . We approximate surplus  $S(\omega)$  by neural network with form:

$$\mathbf{h}^{(1)} = \phi^{(1)}(W^{(1)}\omega + \mathbf{b}^{(1)}) \quad \dots \text{Hidden layer 1}$$

$$\mathbf{h}^{(2)} = \phi^{(2)}(W^{(2)}\mathbf{h}^{(1)} + \mathbf{b}^{(2)}) \quad \dots \text{Hidden layer 2}$$

$$\vdots$$

$$\mathbf{h}^{(H)} = \phi^{(H)}(W^{(H)}\mathbf{h}^{(H-1)} + \mathbf{b}^{(H)}) \quad \dots \text{Hidden layer H}$$

$$S = \sigma(\mathbf{h}^{(H)}) \quad \dots \text{Surplus}$$

- Terminology (our parameter choices are in blue):
  - $H$ : is the number of hidden layers, ( $H = 4$ ; 5 for estimation with pseudo-states)
  - Length of vector  $\mathbf{h}^{(i)}$ : number of neurons in hidden layer  $i$ , ( $Length = 50$ ; 100 for estimation)
  - $\phi^{(i)}$ : is the activation function for hidden layer  $i$ , ( $\phi^i = \tanh$ )
  - $\sigma$ : is the activation function for the final layer, ( $\sigma = \tanh$ )
  - $\Theta = (W^1, \dots, W^{(H)}, b^{(1)}, \dots, b^{(H)})$  are the parameters,

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# Free Entry Condition

---

- Firms make entry decision and then draw type  $y$  from uniform distribution  $[0, 1]$ :

$$0 = \mathbb{E}[V_t^v] = \int V^v(\tilde{y}, z, g) d\tilde{y}. \quad (1)$$

- As the matching function is homothetic  $\frac{m(z_t, g_t)}{\mathcal{V}_t} = \hat{m}\left(\frac{\mathcal{V}_t}{\mathcal{U}_t}\right)$ , combining free entry condition with HJB equation for  $V^v$  gives:

$$\hat{m}\left(\frac{\mathcal{V}_t}{\mathcal{U}_t}\right) = \frac{c}{\int \int \alpha(\tilde{x}, \tilde{y}) \frac{g_t^u(\tilde{x})}{\mathcal{U}_t} (1 - \beta) S_t(\tilde{x}, \tilde{y}) d\tilde{x} d\tilde{y}} \Rightarrow \mathcal{V}_t = \mathcal{U}_t \hat{m}^{-1}(\dots) \quad (2)$$

where  $g_t^u = g_t^w - \int g_t^m(x, y) dy$  and so the RHS can be computed from  $g_t^m$  and  $S_t$ .

- $g_t^f(y) = \mathcal{V}_t + \mathcal{P}_t$  for all  $y$  (uniform), where  $\mathcal{V}_t, \mathcal{P}_t$  can be expressed in terms of  $g$  and  $S$ .
- Timing at a jump in  $z$ :  $z$  changes  $\rightarrow$  exogenous firm exit ( $\sigma$ )  $\rightarrow g_t^f$  adjusts to restore free entry.
- With free entry condition, the master equation expression for surplus takes the same form as without free entry, but with different expressions of  $g^f(y)$ .

# Recursive Equilibrium Part II: Other Equations

---

- Hamilton-Jacobi-Bellman equation (HJBE) for employed worker's value  $V^e(x, y, z, g)$ :

$$\begin{aligned}\rho V^e(x, y, z, g) = & w(x, y, z, g) + \delta(x, y, z) (V^u(x, z, g) - V^e(x, y, z, g)) \\ & + \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z}) (V^e(x, y, \tilde{z}, g^+) - V^e(x, y, z, g)) + D_g V^e(x, y, z, g) \cdot \check{\mu}^g\end{aligned}$$

- HJBE for a vacant firm's value  $V^v(y, z, g)$ :

$$\begin{aligned}\rho V^v(y, z, g) = & -c + \frac{m(z, g)}{\mathcal{V}(z, g)} \int \alpha(\tilde{x}, y, z, g) (V^p(\tilde{x}, y, z, g) - V^v(y, z, g)) \frac{g^u(\tilde{x})}{\mathcal{U}(z, g)} d\tilde{x} \\ & + \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z}) (V^v(y, \tilde{z}, g^+) - V^v(y, z, g)) + D_g V^v(y, z, g) \cdot \check{\mu}^g\end{aligned}$$

- HJBE for a producing firm's value  $V^p(x, y, z, g)$ :

$$\begin{aligned}\rho V^p(x, y, z, g) = & zf(x, y) - w(x, y, z, g) + \delta(x, y, z) (V^v(y, z, g) - V^p(x, y, z, g)) \\ & + \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z}) (V^p(x, y, \tilde{z}, g^+) - V^p(x, y, z, g)) + D_g V^p(x, y, z, g) \cdot \check{\mu}^g\end{aligned}$$

- $g^+ := (1 - \sigma(z, \tilde{z}))g$ : post-jump distribution. No direct  $\varsigma/\sigma$  terms: full compensation for exit.

# Comparison to Other Heterogeneous Agent Search Models

---

- [Lise-Robin '17](#): sets  $\beta = 0$  (and other conditions, including Postal-Vinay Robin style Bertrand competition for workers searching on-the-job)

$$S(x, y, z, g) = S(x, y, z), \quad \alpha(x, y, z, g) = \alpha(x, y, z)$$

- [Menzio-Shi '11](#): competitive search (directed across a collection of sub-markets):

$$S(x, y, z, g) = S(x, y, z)$$

- We look for a solution for  $S$  and  $\alpha$  in terms of the distribution  $g$ .

# Modification 1: Finite Type Approximation

---

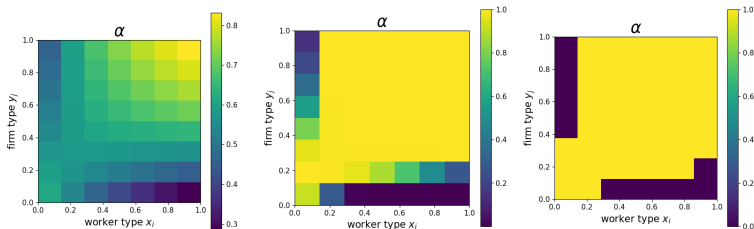
- Approximate  $g(x, y)$  on finite types:  $x \in \mathcal{X} = \{x_1, \dots, x_{n_x}\}$ ,  $y \in \mathcal{Y} = \{y_1, \dots, y_{n_y}\}$ .
- Finite state approximation  $\Rightarrow$  analytical (approximate) KFE:  $g \approx \{g_{ij}\}_{i \leq n_x, j \leq n_y}$
- Approximated master equation for surplus:

$$\begin{aligned} 0 = \mathcal{L}^S S(x, y, z, g) = & -(\rho + \delta(x, y, z))S(x, y, z, g) + zf(x, y) + c - b \\ & - (1 - \beta) \frac{m(z, g)}{\mathcal{V}(z, g)} \frac{1}{n_x} \sum_{i=1}^{n_x} \alpha(\tilde{x}_i, y, z, g) S(\tilde{x}_i, y, z, g) \frac{g^u(\tilde{x}_i)}{\mathcal{U}(z, g)} \\ & - \beta \frac{m(z, g)}{\mathcal{U}(z, g)} \frac{1}{n_y} \sum_{j=1}^{n_y} \alpha(x, \tilde{y}_j, z, g) S(x, \tilde{y}_j, z, g) \frac{g^v(\tilde{y}_j)}{\mathcal{V}(z, g)} \\ & + \sum_{\tilde{z}} \lambda_{z\tilde{z}} (S(x, y, \tilde{z}, g^+) - S(x, y, z, g)) + \sum_{i=1}^{n_x} \sum_{j=1}^{n_y} \partial_{g_{ij}} S(x, y, z, \{g_{ij}\}_{i,j}) \mu^g(\tilde{x}_i, \tilde{y}_j, z, g) \end{aligned}$$

where  $g_{ij}^+ = (1 - \sigma(z, \tilde{z}))g_{ij}$  is the post-jump match distribution.

## Modification 2: Solving the Exact Discrete Choice by Homotopy

- Acceptance  $\alpha$  at steady state  $g$ : smooth ( $\xi = 2$ ,  $\xi = 42$ ) vs. exact discrete rule:



- High curvature ( $\xi = 42$ ) already closely approximates the exact discrete solution.
- Homotopy training is stable: loss declines within each  $\xi$  stage, jumps at stage switches, and the final discrete stage converges from the high- $\xi$  warm start.
- C.f. perturbation approaches (Reiter '09, Bilal '23), which rely on differentiability.

# Finite Dimensional “Distribution” Approximation ( $a = \text{idio. state}$ )

|                                          | Finite Population                                          | Discrete State                                                    | Projection                                                                                                             |
|------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Params $\hat{\varphi}$                   | Agent states<br>$\hat{\varphi}_t = \{(a_t^i)\}_{i \leq N}$ | Masses on grid<br>$\hat{\varphi}_{i,t}, \forall (a^i)_{i \leq N}$ | Basis coefficients<br>$\hat{\varphi}_{i,t}, \forall b_i(a)  _{i \leq N}$                                               |
| Dist. approx.                            | $\frac{1}{N} \sum_{i=1}^N \delta_{(a_t^i)}$                | $\sum_{i=1}^N \hat{\varphi}_{i,t} \delta_{(a^i)}$                 | $\sum_{i=0}^N \hat{\varphi}_{i,t} b_i(a)$<br>$\approx \mu_g \left( a, z, \sum_{i=1}^N \hat{\varphi}_{i,t} b_i \right)$ |
| KFE approx.<br>( $\mu^{\hat{\varphi}}$ ) | Evolution of agents' states                                | Evolution of mass between grid points<br>(e.g. finite diff.)      | Evolution of projection coefficients<br>(least squares)                                                                |

Finite population works well for Walrasian markets; less well for search [Back](#)

# Sampling and Training Scheme for the Surplus Network

---

- Sampling  $(x, y, z)$  is easy (low-dim): multinomial sampler on the discretized support. Sampling the distribution  $(g_{ij})$  is high-dim  $\Rightarrow$  three-phase scheme:
- **Phase 1 — mixed steady-state sampler:**
  1. Solve for the steady state  $g_m^s$  for each fixed aggregate state  $s \in \{L, H, D\}$ .
  2. Draw random convex combinations  $g = \sum_s w_s g_m^s$ : draw a dominant state  $s^*$  with stationary prob.  $\pi_{s^*}$ , set  $w_{s^*} \sim \text{Beta}(\alpha, 1)$  with  $\alpha \gg 1$ , split the rest  $\propto \pi$ .
  3. Train on these samples until the master equation loss falls below a threshold.
- **Phase 2 — ergodic simulation sampler, inner–outer training:**
  1. Simulate  $(z_t, g_t)$  forward with the KFE under the current  $\hat{S}$ ; discard burn-in  $\Rightarrow$  ergodic pool of  $(g, z)$  draws, split into training/validation sets  $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}}$ .
  2. Inner loop: SGD on minibatches from  $\mathcal{D}_{\text{train}}$ ; track validation loss on  $\mathcal{D}_{\text{val}}$  and keep the best weights  $\Theta^*$  (validation-based model selection).
  3. Outer loop: resimulate under  $\Theta^* \Rightarrow$  fresh  $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}}$ ; repeat until validation loss stabilizes.
- **Phase 3 — out-of-sample evaluation:** simulate a new evaluation set  $\mathcal{D}_{\text{eval}}$  after training; report the evaluation loss (the RMSE in our accuracy results).

## Comment 2: Different NN Techniques $\Leftrightarrow$ Different Eqm Concepts

---

Borrowing from Sargent (2008): [Back](#)

- Rational expectations eqm ([this paper](#);  $\mu^g = \check{\mu}^g$ ):
  - Agent “perceived law of motion (PLM)” is the true law of motion if agents follow PLM
  - True on and off the equilibrium path  $\Rightarrow$  requires solving the master equation.
- Self-confirming eqm ([some model-based reinforcement learning, e.g. DeepHAM](#))
  - PLM is best statistical fit to data generated by economy where agents follow that PLM.
  - So, agents would never reject a misspecification test
  - but may have off-equilibrium beliefs that differ from rational expectations ( $\mu^g \neq \check{\mu}^g$ )
  - $\Rightarrow$  can be solved by updating reaction functions.
- Misspecified belief eqm
  - Agents have a “perceived law of motion” for the equilibrium variables in the economy
  - ...that satisfies a fixed parametric form but doesn't satisfy a misspecification test.

# Calibration and Estimation Challenges

---

- Let  $\Psi \in \Omega^\Psi$  be the structural parameters to be calibrated internally.
- Let  $\check{\varphi} = (\check{\varphi}_1, \dots, \check{\varphi}_N)$  be the  $N \times 1$  data moments that we want to match.
- Let  $\phi(\Psi) = (\phi_1(\Psi), \dots, \phi_N(\Psi))$  be the corresponding simulated model moments.
- Computation challenge: need to solve  $\phi(\Psi)$  for many structural parameters  $\Psi$ .
- Solution: include  $\Psi$  as a pseudo state vector and solve extended master equation:

$$0 = \mathcal{L}^S S(x, y, z, \underline{g}, \Psi)$$

- Approximate extended surplus function by NN:  $S(x, y, z, g, \Psi) \approx \hat{S}(x, y, z, g, \Psi; \Theta)$ .
- Fit  $\hat{S}$  using same approach as before but sampling over states & structural parameters.

# Simulated Method of Moments

---

- Fit surrogate neural network  $\hat{\Phi}$  mapping structural parameters to simulated moments:

$$\Psi \mapsto \hat{\Phi}(\Psi; \Theta^\Phi) = (\hat{\phi}_1(\Psi; \Theta^\Phi), \dots, \hat{\phi}_N(\Psi; \Theta^\Phi)), \quad \Theta^\Phi \text{ are NN parameters}$$

- Simulate the model under many different structural parameter vectors  $\{\Psi_I\}_I$
  - Compute the resulting model moments  $\{(\phi_1(\Psi_I), \dots, \phi_N(\Psi_I))\}_I$
  - Fit neural network parameters  $\Theta^\Phi$  to approx relationship  $\hat{\Phi}(\Psi; \Theta^\Phi)$  using simulations.
- Think: surrogate NN has incorporated the economic structure implicitly into  $\Theta^\Phi$
  - With  $\hat{\Phi}(\Psi; \Theta^\Phi)$ , we can find the parameters to match data moments:  
(i.e. perform simulated method of moments)

$$\Psi^* = \arg \min_{\Psi} \sum_{i=1}^N \omega_i \left( \frac{\check{\varphi}_i - \hat{\phi}_i(\Psi; \Theta^\Phi)}{\check{\varphi}_i} \right)^2.$$

# Calibration and Estimation

---

1. Fit extended surplus function by NN:  $S(x, y, z, g, \Psi) \approx \hat{S}(x, y, z, g, \Psi; \Theta)$ .

Start with initial parameter guess  $\Theta^0$ . At iteration  $n$  with  $\Theta^n$ :

- 1.1 Generate  $K$  sample points,  $Q^n = \{(x_k, y_k, z_k, \{g_{ij,k}\}_{i \leq n_x, j \leq n_y}, \Psi_k)\}_{k \leq K}$ .

- 1.2 Calculate the average mean squared error of surplus master equation on sample points:

$$L(\Theta^n, Q^n) := \frac{1}{K} \sum_{k \leq K} \left| \mathcal{L}^S \hat{S}(x_k, y_k, z_k, \{g_{ij,k}\}_{i \leq n_x, j \leq n_y}, \Psi_k) \right|^2$$

- 1.3 Update NN parameters with stochastic gradient descent (SGD) method.

2. Fit surrogate NN mapping structural parameters to moments:  $\hat{\Phi}(\Psi; \Theta^\Phi)$ .

3. Use surrogate NN to find parameters that match data moments.

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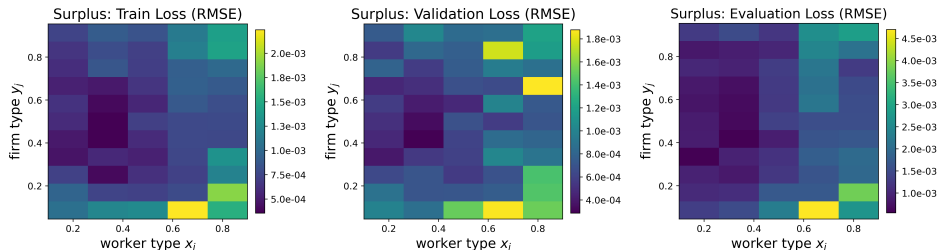
On-The-Job Search Model with Idiosyncratic Shocks

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# Numerical Performance: Accuracy I [back](#)

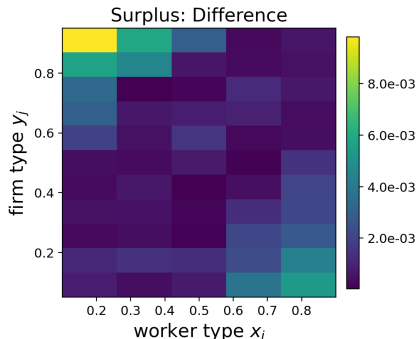
- RMSE of the surplus master equation over the ergodic mean, computed on: (a) training samples (last inner loop), (b) validation samples (last outer loop), (c) a fresh simulated evaluation set.



- Average RMSEs:  $9.2 \times 10^{-4}$  (training),  $9.1 \times 10^{-4}$  (validation),  $1.7 \times 10^{-3}$  (evaluation)  
 $\Rightarrow$  accurate approximation of  $S$  in the neighborhood of the ergodic mean.

# Numerical Performance: Accuracy II

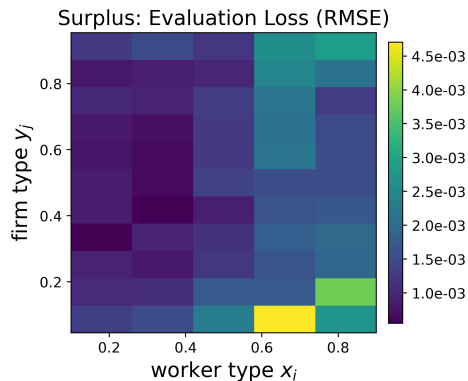
- Compare **steady state solution without aggregate shocks** to solution using conventional methods.



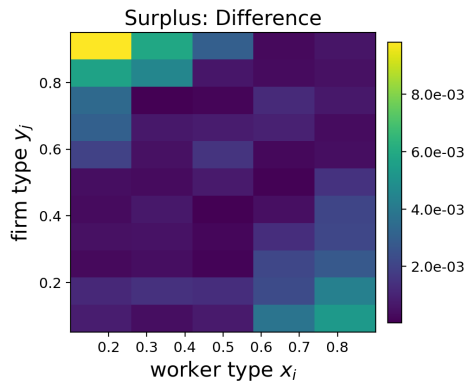
**Figure:** Absolute difference of  $S$  from conventional solution at DSS (order  $10^{-3}$ ; max  $S \approx 0.65$ ).

# Calibration of Model with Aggregate Shocks Back

| Parameter                               | Interpretation                    | Value                                        | Target/Source                              |
|-----------------------------------------|-----------------------------------|----------------------------------------------|--------------------------------------------|
| $\rho$                                  | Discount rate                     | 0.05                                         | Interest rate                              |
| $\xi$                                   | Extreme value for $\alpha$ choice | 2.0                                          |                                            |
| $f(x, y)$                               | Production for match $(x, y)$     | $0.6 + 0.4 (\sqrt{x} + \sqrt{y})^2$          | Hagedorn et al. (2017)                     |
| $\beta$                                 | Surplus division factor           | 0.72                                         | Shimer (2005)                              |
| $m(\mathcal{U}, \mathcal{V})$           | Matching function                 | $\kappa \mathcal{U}^\nu \mathcal{V}^{1-\nu}$ | Hagedorn et al. (2017)                     |
| $\nu$                                   | Elasticity in meeting function    | 0.5                                          | Hagedorn et al. (2017)                     |
| $\kappa$                                | Scale for meeting function        | 6.4                                          | Unemployment rate                          |
| $b$                                     | Worker unemployment benefit       | 1.0                                          | Unemployment rate                          |
| $c$                                     | Entry cost                        | 2.40                                         | Steady state $\mathcal{V}/\mathcal{U} = 1$ |
| Steady State:                           |                                   |                                              |                                            |
| $\bar{z}$                               | Steady state TFP                  | 1                                            | Shimer (2005)                              |
| $\bar{\delta}$                          | Steady state separation rate      | 0.2                                          | BLS job tenure 5 years                     |
| $\bar{\varsigma}$                       | Steady state exogenous exit rate  | 0.01                                         | No endogenous exit                         |
| Exogenous Aggregate Shock Process:      |                                   |                                              |                                            |
| $A$                                     | TFP level                         | 1.0                                          |                                            |
| $\varsigma_D, \varsigma_L, \varsigma_H$ | Exogenous exit rates              | 0.63, 0.01, 0.01                             | No endogenous exit                         |
| $\delta_L, \delta_H$                    | Separation rates                  | 0.22, 0.18                                   | Shimer (2005)                              |
| $\delta_D(x, y)$                        | Separation at crisis state        | 0.23 to 3.9                                  | Cajner et al. (2020)                       |
| $\lambda_z$                             | Poisson transition probabilities  | 0.4, 0.001, 0.0995                           | Shimer (2005)                              |
| $\sigma(z, \check{z})$                  | Exogenous exit at Poisson jumps   | 0 to 0.05                                    | No endogenous exit                         |
| $n_x$                                   | Discretization of worker types    | 5                                            | Cajner et al. (2020)                       |
| $n_y$                                   | Discretization of firm types      | 11                                           | Cajner et al. (2020)                       |



(a) Model with aggregate shock:  
evaluation RMSE across the  $(x, y)$  grid



(b) Model without aggregate shock:  
difference from conventional solution

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# Different Versions of Entry and Exit

---

1. Baseline (main text): free entry + exogenous exit at rate  $\varsigma(z)$  and fraction  $\sigma(z, \check{z})$  at  $z$  jumps, with full compensation for exit ( $g^f$  is a jump variable).  
Calibrated s.t.  $\varsigma, \sigma$  large enough  $\Rightarrow$  no endogenous exit on equilibrium path.
2. General case: endogenous firm exit + partial compensation (fraction  $\tau \in [0, 1]$  of lost surplus) ( $g^f$  jump variable).  $\tau = 1$  & no endogenous exit  $\Rightarrow$  baseline.
3. Short-lived vacancies (as in Lise-Robin '17; used in our OJS model): unfilled vacancies destroyed each instant,  $V^v = 0$ ; no compensation;  $\varsigma = \sigma = 0$ .
4. Slow entry and exit (mass of firms  $\mathcal{F}$  is a state variable).

# 1. Baseline: Free Entry, Exogenous Compensated Exit Back

- KFE: drift includes firm exit at rate  $\varsigma(z)$ ; exogenous jump exit when  $z$  changes:

$$dg_t(x, y) = \mu^g(x, y, z_t, g_t) dt - \sigma(z_t, \tilde{z}) g_t(x, y) dN_t(z_t; \tilde{z})$$

$$\mu^g(x, y, z, g) = -(\delta(x, y, z) + \varsigma(z))g(x, y) + \frac{m(z, g)}{\mathcal{U}(z, g)\mathcal{V}(z, g)}\alpha(x, y, z, g)g^v(y)g^u(x)$$

- Master equation: no direct  $\varsigma/\sigma$  loss terms (exit compensated); jump at post-jump dist.:

$$\begin{aligned} \rho S(x, y, z, g) = & zf(x, y) - \delta(x, y, z)S(x, y, z, g) + c - b \\ & - (1 - \beta) \frac{m(z, g)}{\mathcal{V}(z, g; S)} \int \alpha(\tilde{x}, y, z, g) S(\tilde{x}, y, z, g) \frac{g^u(\tilde{x})}{\mathcal{U}(z, g)} d\tilde{x} \\ & - \beta \frac{m(z, g)}{\mathcal{U}(z, g)} \int \alpha(x, \tilde{y}, z, g) S(x, \tilde{y}, z, g) \frac{g^v(\tilde{y})}{\mathcal{V}(z, g; S)} d\tilde{y} \\ & + \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z}) (S(x, y, \tilde{z}, (1 - \sigma(z, \tilde{z}))g) - S(x, y, z, g)) + D_g S \cdot \mu^g \end{aligned}$$

- Firm distribution  $g^f$  jumps to ensure free entry condition is satisfied:

$$0 = \mathbb{E}[V_t^v] = \int V^v(\tilde{y}, z, g) d\tilde{y}.$$

## 2. General Case: Endogenous Exit and Partial Compensation Back

- If free entry would require negative entry, firms are randomly forced to exit: at rate  $\alpha_\mu^g(z, g)$  in the drift (indicator  $\mathbf{1}_{\mu^F}$ ) or with prob.  $\alpha_\sigma^g(z, \check{z}, g)$  at  $z$  jumps (indicator  $\mathbf{1}_{\sigma^F}$ ), chosen so the free entry condition keeps holding.
- Exiting firms (workers) receive a transfer for fraction  $\tau$  of their lost surplus.
- Master equation with endogenous exit and partial compensation:

$$\begin{aligned}
 \rho S(x, y, z, g) = & z f(x, y) - \delta(x, y, z) S(x, y, z, g) - (\varsigma(z) + \mathbf{1}_{\mu^F} \alpha_\mu^g)(1 - \tau) S(x, y, z, g) \\
 & + c - (1 - \beta) \frac{m(z, g)}{\mathcal{V}(z, g; S)} \int \alpha(\tilde{x}, y, z, g) S(\tilde{x}, y, z, g) \frac{g^u(\tilde{x})}{\mathcal{U}(z, g)} d\tilde{x} \\
 & - b - \beta \frac{m(z, g)}{\mathcal{U}(z, g)} \int \alpha(x, \tilde{y}, z, g) S(x, \tilde{y}, z, g) \frac{g^v(\tilde{y})}{\mathcal{V}(z, g; S)} d\tilde{y} \\
 & + \sum_{\check{z} \neq z} \lambda(z, \check{z}) \left\{ S(x, y, \check{z}, g + \gamma^g) - S(x, y, z, g) \right. \\
 & \left. - ((1 - \sigma) \mathbf{1}_{\sigma^F} \alpha_\sigma^g + \sigma)(1 - \tau) S(x, y, \check{z}, g + \gamma^g) \right\} + \langle D_g S, \mu^g \rangle
 \end{aligned}$$

where  $\gamma^g = -(\sigma + \mathbf{1}_{\sigma^F} \alpha_\sigma^g)g$  is the jump in  $g$ ;  $\mu^g$  includes exit rate  $\varsigma + \mathbf{1}_{\mu^F} \alpha_\mu^g$ .

- $\tau = 1$  and  $\mathbf{1}_{\mu^F} = \mathbf{1}_{\sigma^F} = 0$  recovers the baseline master equation.

### 3. Slow Entry and Exit Back

- Vacancies long-lived; firms exit at exogenous rate  $\varsigma$  (no jump exit,  $\sigma = 0$ ); entrants draw type  $y$  from density  $\bar{h}^f(y) = 1$ ; entry at rate  $\psi \alpha^f$ :

$$dg_t^f(y) = \underbrace{\psi \alpha^f(z, g, \mathcal{F}) \bar{h}^f(y)}_{\text{entry inflow}} dt - \underbrace{\varsigma g_t^f(y)}_{\text{exit outflow}} dt$$

$$\alpha^f(z, g, \mathcal{F}) = \mathbb{1} \left\{ \int V^v(\tilde{y}, z, g, \mathcal{F}) \bar{h}^f(\tilde{y}) d\tilde{y} > 0 \right\}$$

- Firm density stays uniform  $\Rightarrow$  only the mass of firms  $\mathcal{F}_t$  is an extra state:

$$d\mathcal{F}_t = \psi \alpha^f(z, g, \mathcal{F}) dt - \varsigma \mathcal{F}_t dt, \quad \mathcal{V}_t = \mathcal{F}_t - \mathcal{P}_t$$

- Surplus function has an additional state (and we need to solve for  $V^v$ ):  $S(x, y, z, g, \mathcal{F})$ , with the master equation gaining the term  $\partial_{\mathcal{F}} S(x, y, z, g, \mathcal{F}) \mu^{\mathcal{F}}(z, g, \mathcal{F})$ .
- Limits:  $\psi \rightarrow \infty$  recovers free entry ( $g^f$  jump variable);  $\psi \rightarrow 0$  shuts down entry.

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# On-The-Job Search (OJS): Additional Environment Features

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- Same worker types, firm types, and production function.
- Now all workers search; meetings  $m(\mathcal{W}_t, \mathcal{V}_t)$ ; total search effort  $\mathcal{W}_t := \mathcal{U}_t + \phi \mathcal{E}_t$ .  
Meeting rates:  $\mathcal{M}_t^u = \frac{m_t}{\mathcal{W}_t}$  (unemployed),  $\mathcal{M}_t^e = \phi \frac{m_t}{\mathcal{W}_t}$  (employed),  $\mathcal{M}_t^v = \frac{m_t}{\mathcal{V}_t}$  (vacancy).
- Vacant  $\tilde{y}$ -firm meets unemployed  $x$ -worker: Nash bargaining with worker weight  $\beta$ .
- Poaching firm  $\tilde{y}$  meets worker in  $(x, y)$  match: [parameterized bargaining protocol](#):
  - Worker moves iff  $S_t(x, \tilde{y}) \geq S_t(x, y) \geq 0$ ; gets share  $\beta$  of incremental surplus  $S_t(x, \tilde{y}) - S_t(x, y)$  ... and share  $\chi \in [0, 1]$  of the incumbent firm's surplus.
  - Nests: Bertrand competition ( $\chi = 1, \beta = 0$ ; PVR '02), sequential auctions ( $\chi = 1, \beta > 0$ ; CPVR '06), incremental Nash bargaining ( $\chi = 0$ ; Pissarides '94).
  - $\chi$  does not enter the surplus master equation (only wages)  $\Rightarrow$  same  $S$  for all  $\chi$ .
- Endogenous separation at rate  $\eta$  when  $S_t(x, y) < 0$ :  $\alpha_t^b(x, y) = 1 - \alpha_t(x, y)$ .
- Short-lived vacancies (Lise-Robin '17): unfilled vacancies destroyed each instant  $\Rightarrow V^v = 0$ ; free entry on the flow value of a vacancy;  $\varsigma = \sigma = 0$ .

# Recursive Characterization For Equilibrium Surplus (OJS)

---

- Can characterize equilibrium with the master equation for the surplus:

$$\begin{aligned} \rho S(x, y, z, g) = & z f(x, y) - b - (\delta(x, y, z) + \eta \alpha^b(x, y, z, g)) S(x, y, z, g) \\ & + \mathcal{M}^e \int \alpha^e(x, y, \tilde{y}, z, g) \beta(S(x, \tilde{y}, z, g) - S(x, y, z, g)) \frac{g^v(\tilde{y})}{\nu} d\tilde{y} \\ & - \mathcal{M}^u \int \alpha(x, \tilde{y}, z, g) \beta S(x, \tilde{y}, z, g) \frac{g^v(\tilde{y})}{\nu} d\tilde{y} \\ & + \sum_{\tilde{z} \neq z} \lambda(z, \tilde{z}) (S(x, y, \tilde{z}, g) - S(x, y, z, g)) + D_g S(x, y, z, g) \cdot \mu^g(z, g) \end{aligned}$$

where the on-the-job acceptance and endogenous breakup indicators are:

$$\alpha^e(x, y, \tilde{y}, z, g) := \mathbb{1}\{S(x, \tilde{y}, z, g) \geq S(x, y, z, g) \text{ and } S(x, \tilde{y}, z, g) \geq 0\}, \quad \alpha^b = 1 - \alpha$$

- Simpler than baseline:  $V^v = 0$  (short-lived vacancies)  $\Rightarrow$  no firm opportunity cost term.  
 $\beta = 0$  kills both  $g$ -dependent terms  $\Rightarrow$  block recursivity as in Lise-Robin '17.

## OJS: Free Entry Condition

---

- The flow value of creating a vacancy  $y$ : rents from hiring unemployed workers and poaching employed workers ( $\mathcal{C}^u = \frac{\mathcal{U}}{\mathcal{W}}$ ,  $\mathcal{C}^e = \frac{\phi \mathcal{E}}{\mathcal{W}}$  prob. met worker is u/e):

$$\begin{aligned} J^v(y, z, g) = & \mathcal{M}^v \mathcal{C}^u \int \alpha(\tilde{x}, y, z, g) (1 - \beta) S(\tilde{x}, y, z, g) \frac{g^u(\tilde{x})}{\mathcal{U}} d\tilde{x} \\ & + \mathcal{M}^v \mathcal{C}^e \int \int \alpha^e(\tilde{x}, \tilde{y}, y) (1 - \beta) [S(\tilde{x}, y, z, g) - S(\tilde{x}, \tilde{y}, z, g)] \frac{g(\tilde{x}, \tilde{y})}{\mathcal{E}} d\tilde{x} d\tilde{y} \end{aligned}$$

- Free entry condition (entrants draw  $y$  from uniform density  $\bar{h}^v(\tilde{y}) = 1$ ):

$$0 = -c + \int_0^1 J^v(\tilde{y}, z, g) \bar{h}^v(\tilde{y}) d\tilde{y}$$

- With homothetic  $m(\mathcal{W}, \mathcal{V})$ , invert to solve  $\mathcal{V}$  as a function of  $(z, g)$  and  $S(\cdot, \cdot, z, g)$ .

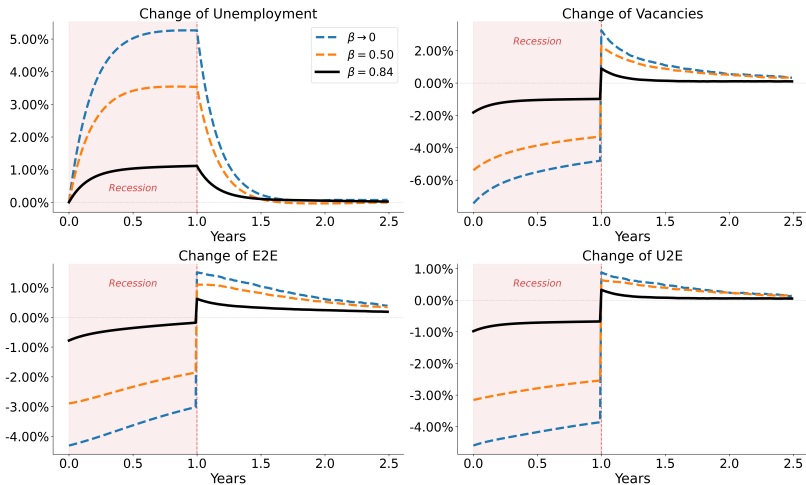
# On-the-job-search: KFE

---

- The KFE becomes:

$$\begin{aligned}
 dg_t(x, y) = & \underbrace{-(\delta(x, y, z) + \eta \alpha^b(x, y, z_t, g_t)) g_t(x, y) dt}_{\text{Outflow: exogenous \& endogenous separation}} \\
 & - \underbrace{\mathcal{M}_t^e \int \alpha^e(x, y, \tilde{y}, z_t, g_t) \frac{g_t^v(\tilde{y})}{\mathcal{V}_t} d\tilde{y} g_t(x, y) dt}_{\text{Outflow: workers poached to better matches}} \\
 & + \underbrace{\mathcal{M}_t^u \alpha(x, y, z_t, g_t) \frac{g_t^v(y)}{\mathcal{V}_t} g_t^u(x) dt}_{\text{Inflow: unemployed workers hired}} \\
 & + \underbrace{\mathcal{M}_t^e \int \alpha^e(x, \tilde{y}, y, z_t, g_t) \frac{g_t^v(y)}{\mathcal{V}_t} g_t(x, \tilde{y}) d\tilde{y} dt}_{\text{Inflow: workers poached from other matches}} =: \mu^g(x, y, z_t, g_t) dt
 \end{aligned}$$

## Q4. Block recursive model impact on IRF to negative TFP shock?



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# On-The-Job Search Model with Idiosyncratic Shocks

---

- Introduce match specific (idiosyncratic) productivity shock  $\epsilon$  (Mortensen and Pissarides, 1994) that follows a continuous time Markov chain with intensities  $\lambda(\epsilon, \epsilon')$ .
- Production output from matches  $F(x, y, \epsilon, z) = A_z A_\epsilon f(x, y)$ ; new matches draw initial  $\epsilon \sim \pi(\epsilon)$ .
- **Endogenous separation**: agents immediately break up the match when a new draw of  $\epsilon$  makes the surplus negative ( $\alpha^b(x, y, \epsilon, z, g) = \mathbb{1}\{S(x, y, \epsilon, z, g) < 0\}$ ).

# On-The-Job Search Model with Idiosyncratic Shocks: Master Equation

---

$$\begin{aligned}
 \rho S(x, y, \epsilon, z, g) &= \rho(P(x, y, \epsilon, z, g) - V^u(x, z, g)), \quad V^v = 0 \text{ (short-lived vacancies)} \\
 &= F(x, y, \epsilon, z) - \delta(x, y, z)S(x, y, \epsilon, z, g) - b \\
 &+ \mathcal{M}^e \sum_{\tilde{\epsilon}} \pi(\tilde{\epsilon}) \int \alpha^e(x, y, \epsilon, \tilde{y}, \tilde{\epsilon}, z, g) \beta(S(x, \tilde{y}, \tilde{\epsilon}, z, g) - S(x, y, \epsilon, z, g)) \frac{g^v(\tilde{y})}{\nu} d\tilde{y} \\
 &- \mathcal{M}^u \sum_{\tilde{\epsilon}} \pi(\tilde{\epsilon}) \int \alpha(x, \tilde{y}, \tilde{\epsilon}, z, g) \beta S(x, \tilde{y}, \tilde{\epsilon}, z, g) \frac{g^v(\tilde{y})}{\nu} d\tilde{y} \\
 &+ \sum_{\check{\epsilon} \neq \epsilon} \lambda(\epsilon, \check{\epsilon}) ((1 - \alpha^b(x, y, \check{\epsilon}, z, g))(S(x, y, \check{\epsilon}, z, g) - S(x, y, \epsilon, z, g)) \\
 &\quad - \alpha^b(x, y, \check{\epsilon}, z, g) S(x, y, \epsilon, z, g)) \\
 &+ \sum_{\check{z} \neq z} \lambda(z, \check{z}) (S(x, y, \epsilon, \check{z}, g) - S(x, y, \epsilon, z, g)) + \langle D_g S, \mu^g \rangle
 \end{aligned}$$

where  $\alpha^e(x, y, \epsilon, \tilde{y}, \tilde{\epsilon}, z, g) = \mathbb{1}\{S(x, \tilde{y}, \tilde{\epsilon}, z, g) \geq S(x, y, \epsilon, z, g) \text{ \& } S(x, \tilde{y}, \tilde{\epsilon}, z, g) \geq 0\}$ .

# On-The-Job Search Model with Idiosyncratic Shocks: KFE

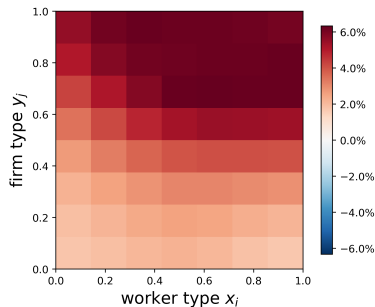
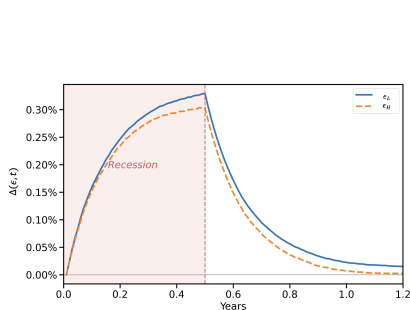
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$$\begin{aligned}
 dg_t(x, y, \epsilon) = & -\delta(x, y, z) g_t(x, y, \epsilon) dt \\
 & - \underbrace{\underbrace{\mathcal{M}_t^e}_{\text{Rate. e-worker meets}} \sum_{\tilde{\epsilon}} \pi(\tilde{\epsilon}) \int \underbrace{\alpha_t^e(x, y, \epsilon, \tilde{y}, \tilde{\epsilon})}_{\text{Prob. accept}} \underbrace{\frac{g_t^v(\tilde{y})}{\mathcal{V}_t}}_{\text{Prob. meet } \tilde{y}} d\tilde{y}}_{\text{Mass accepting poaching offer}} \underbrace{g_t(x, y, \epsilon)}_{\text{Mass at } (x, y, \epsilon)} dt \\
 & + \underbrace{\mathcal{M}_t^u}_{\text{Rate. u-worker meets}} \pi(\epsilon) \alpha_t(x, y, \epsilon) \frac{g_t^v(y)}{\mathcal{V}_t} g_t^u(x) dt \\
 & + \mathcal{M}_t^e \pi(\epsilon) \sum_{\tilde{\epsilon}} \int \alpha_t^e(x, \tilde{y}, \tilde{\epsilon}, y, \epsilon) \frac{g_t^v(y)}{\mathcal{V}_t} g_t(x, \tilde{y}, \tilde{\epsilon}) d\tilde{y} dt \\
 & - \sum_{\check{\epsilon} \neq \epsilon} \lambda(\epsilon, \check{\epsilon}) g_t(x, y, \epsilon) dt + \sum_{\check{\epsilon} \neq \epsilon} \lambda(\check{\epsilon}, \epsilon) (1 - \alpha_t^b(x, y, \epsilon)) g_t(x, y, \check{\epsilon}) dt
 \end{aligned}$$

(Last line: outflow when  $\epsilon$  switches away; inflow from  $(x, y, \check{\epsilon})$  when  $\check{\epsilon}$  switches to  $\epsilon$ , net of endogenous breakups.)

# On-The-Job Search Model with Idiosyncratic Shocks: Results

- Recession experiment ( $z = z_L$  for one year): cumulative net outflows from matches rise more, and recover more slowly, at the low idiosyncratic state  $\epsilon_L$  (left)  
...because low- $\epsilon$  matches carry lower surplus and sit near the breakup threshold  $S = 0$ .
- Endogenous separation flow  $Q(x, y, z, g_{\text{erg}})$ : recessions raise endogenous separations relatively more for high- $(x, y)$  matches (right).



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# Environment: Setting, Bonds, and Households

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- Continuous time, infinite horizon environment.
- There are many bonds,  $k \in \{1, \dots, K\}$ , in positive net supply  $s_k$ :
  - Every bond pays the same dividend  $\delta > 0$ .
  - Bond  $k$  matures at rate  $1/\tau_k$  (average maturity  $\tau_k$ ).
- Populated by a unit-mass continuum of infinitely-lived and risk-neutral investors:
  - An investor can hold either zero or one share of at most one type of asset.
  - Investor type  $j \in \{1, \dots, J\}$  gets flow utility  $\delta - \psi(j, k)$  from holding bond  $k$ .
  - Agents **switch type** from  $i$  to  $j$  at rate  $\lambda_{i,j}$ .
- Aggregate (default) state  $z \in \{z_1, \dots, z_n\}$ , switches at rate  $\zeta_{z,z'}$ .  
At state  $z$ , asset  $k$  pays a fraction  $\phi(k, z)$  of the coupon and the principal.

# Distribution and Bargaining

---

- An investor's state is made up of her holding cost  $j \in \{1, \dots, J\}$  and her ownership status, for each asset type  $k \in \{1, \dots, K\}$  (owner  $o$  or non-owner  $n$ ). Hence the set of investor idiosyncratic states is:

$$A = \{1n, 2n, \dots, Jn, 1o1, \dots, 1oK, 2o1, \dots, 2oK, Jo1, \dots, JoK\} \quad (3)$$

- The rate of contact between investors with states  $a$  and  $b$  is:

$$\mathcal{M}_{a,b} = \kappa_{a,b} g_a g_b \quad (4)$$

- Agents  $a, b$  **trade with each other**, engage in generalized Nash bargaining with bargaining power  $\beta_{a,b}$ .

# Value Functions for Non-Owners and Owners [back](#)

- Value function for non-owner with type  $i$  satisfies:

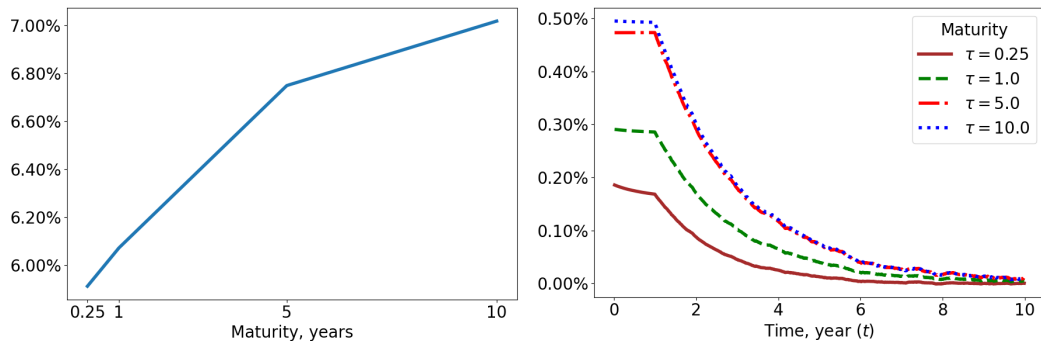
$$\begin{aligned}\rho_i V(in, g, z) &= \sum_a \kappa_{in,a} \alpha(in, a, g, z) \beta_{in,a} S(in, a, z, g) \\ &+ \sum_k \xi_{i,k} (V(iok, g, z) - V(in, g, z)) + \sum_{j \neq i} \lambda_{i,j} (V(jn, g, z) - V(in, g, z)) \\ &+ \sum_{z'} \zeta_{z,z'} (V(in, g, z') - V(in, g, z)) + \sum_{a \in A} \partial_{g_a} V(in, g, z) \mu^g(a, z)\end{aligned}$$

- Value function for an investor of type  $i$  holding asset  $k$  satisfies:

$$\begin{aligned}\rho_i V(iok, g, z) &= \delta \phi(k, z) - \psi(i, k) + \frac{1}{\tau_k} (V(in, g, z) + \pi(k, z) - V(iok, g, z)) \\ &+ \sum_a \kappa_{iok,a} \alpha(iok, a, g, z) g_a \beta_{iok,a} S(iok, a, g, z) + \sum_{j \neq i} \lambda_{i,j} (V(jok, g, z) - V(iok, g, z)) \\ &+ \sum_{z'} \zeta_{z,z'} (V(iok, g, z') - V(iok, g, z)) + \sum_{a \in A} \partial_{g_a} V(iok, g, z) \mu^g(a, z).\end{aligned}$$

$\alpha(in, jok, g, z)$ : indicator for whether the trade is accepted upon matching.

# Endogenous Yield Curve and Impulse Responses

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**Figure:** Yield curve at ergodic steady state and impulse responses. Right figure: proportional bond yield change compared to the ergodic yield at each maturity following a one-year recession.

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# Non-transferable Utility (NTU)

---

- Main change compared to the benchmark model: for a match  $(x, y)$ , agent  $x$  receives payoff  $zf(x, y)$  and agent  $y$  receives  $zf(y, x)$ . No surplus to be split via bargaining.
- Computational implication:
  1. We no longer solve a single master equation for the match surplus.
  2. Instead, we directly approximate the value functions  $V^u(x, z, g)$ ,  $V^e(x, y, z, g)$ ,  $V^v(y, z, g)$ , and  $V^p(x, y, z, g)$  using four neural networks, trained jointly to minimize a weighted sum of residuals from the four master equations.

# Non-transferable Utility (NTU, Ct'd)

- Mean squared loss as a function of type in the master equations of  $V^u$  (left) and  $V^e$  (right)

